

Virtual homological torsion in low dimensions

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University of Bonn

Overview

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- Virtual homological torsion: an introduction

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- A 2-dimensional setting

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- What's next?

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Conjecture (Bergeron–Venkatesh)

Let M be a closed, hyperbolic 3-manifold and let

$$M = M_0 \leftarrow M_1 \leftarrow M_2 \leftarrow \cdots$$

be a cofinal tower of finite-sheeted normal covers of M , that is,

$$\bigcap_n \pi_1(M_n) = 1.$$

Then

$$\lim_{n \rightarrow \infty} \frac{\log(|\mathrm{Tor}(H_1(M_n; \mathbb{Z}))|)}{[\pi_1(M) : \pi_1(M_n)]} = \frac{\mathrm{vol}(M)}{6\pi}.$$



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BV Conjecture: $\forall p, \quad \lim_{n \rightarrow \infty} \frac{\log(|\text{Tor}_p(H_1(M_n; \mathbb{Z}))|)}{[\pi_1(M) : \pi_1(M_n)]} = 0,$

so torsion must involve larger and larger sporadic primes.

Open problem

Is there a finitely presented residually finite group G , and a residual normal chain

$$G \supseteq G_1 \supseteq G_2 \supseteq \cdots$$

(i.e. $\bigcap_n G_n = \{1\}$), such that

$$\lim_{n \rightarrow \infty} \frac{\log(|\mathrm{Tor}(G_n^{\mathrm{ab}})|)}{[G : G_n]} > 0 ?$$

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- **Finite presentation:** (Kar–Kropholler–Nikolov '17) For every $f : \mathbb{N} \rightarrow \mathbb{N}$ there is a fg G and an exhausting normal chain $G \supseteq G_1 \supseteq G_2 \supseteq \cdots$ such that $|\mathrm{Tor}(G^{\mathrm{ab}})| > f([G : G_n])$.

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- **Normality:** (Liu '19) Cofinal towers of finite-sheeted *non-normal* covers of closed, hyperbolic M^3 with exponential growth.

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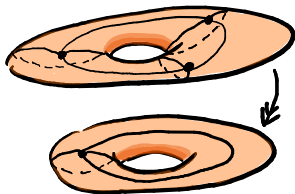
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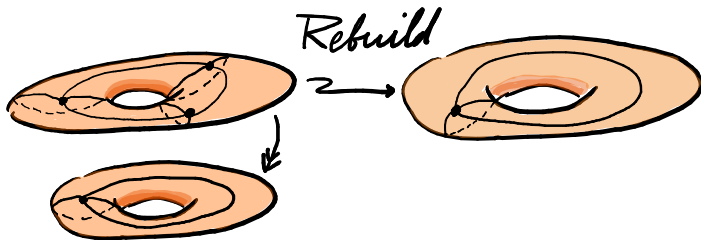
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Back to 3-manifolds

Theorem (Sun '15)

Let M be a closed hyperbolic 3-manifold and let A be a finite abelian group. There exists a finite-sheeted cover $\widehat{M} \rightarrow M$ such that

$$H_1(\widehat{M}; \mathbb{Z}) = A \oplus \cdots .$$

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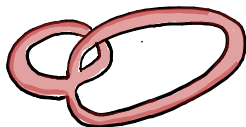
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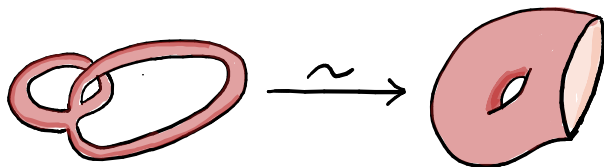
Similar proof works for all 3-manifolds with empty or toroidal boundary which are not graph manifolds (Chu–Groves '22).

Virtual homological torsion in 3-manifolds: a recipe

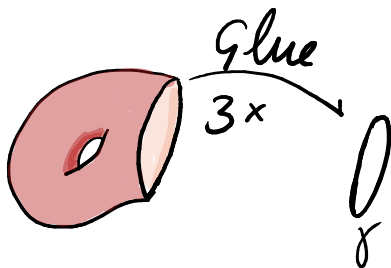
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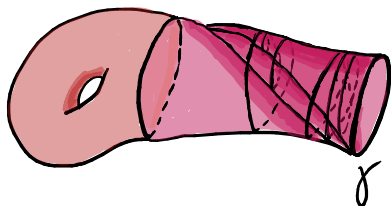
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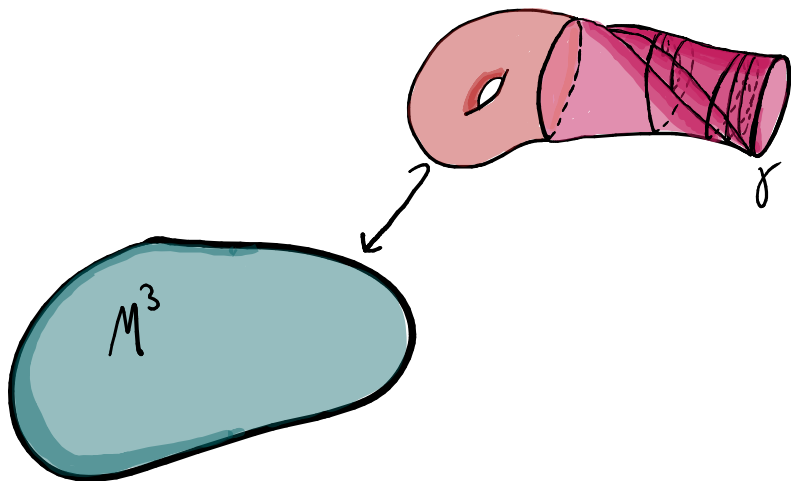
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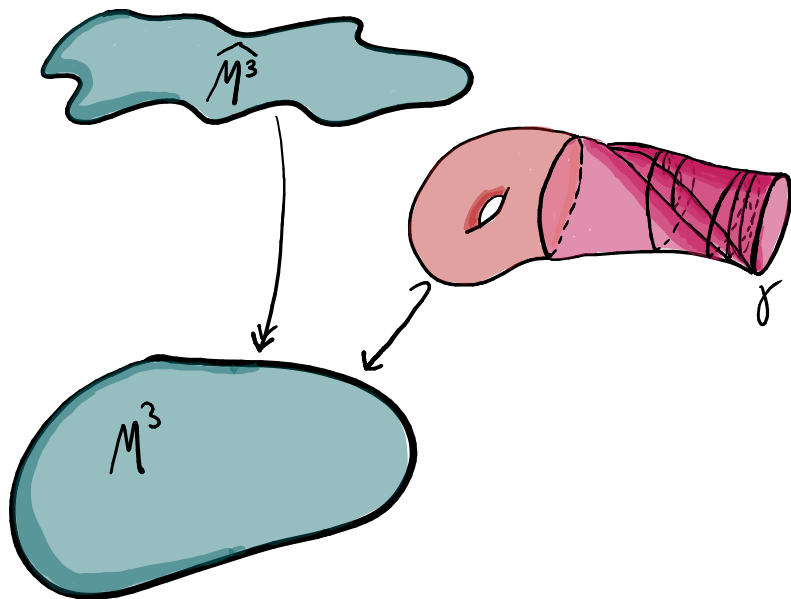
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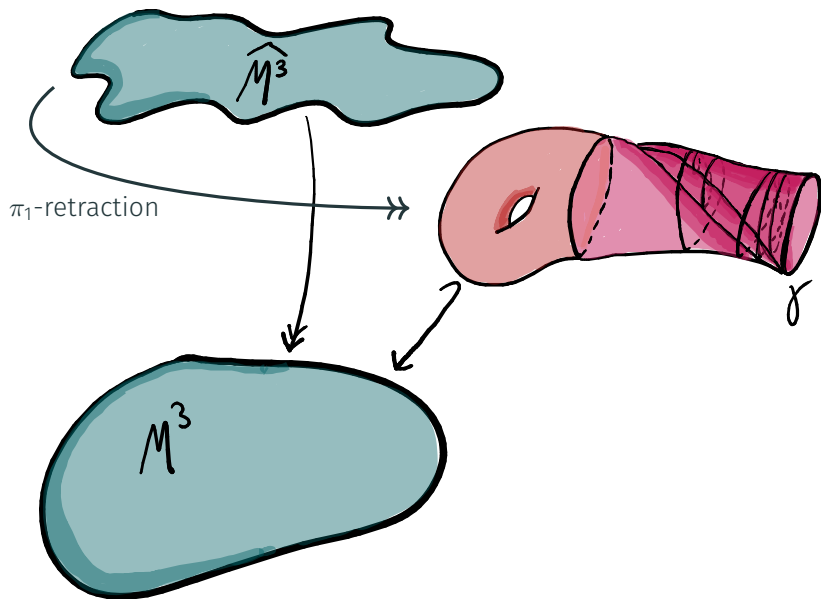
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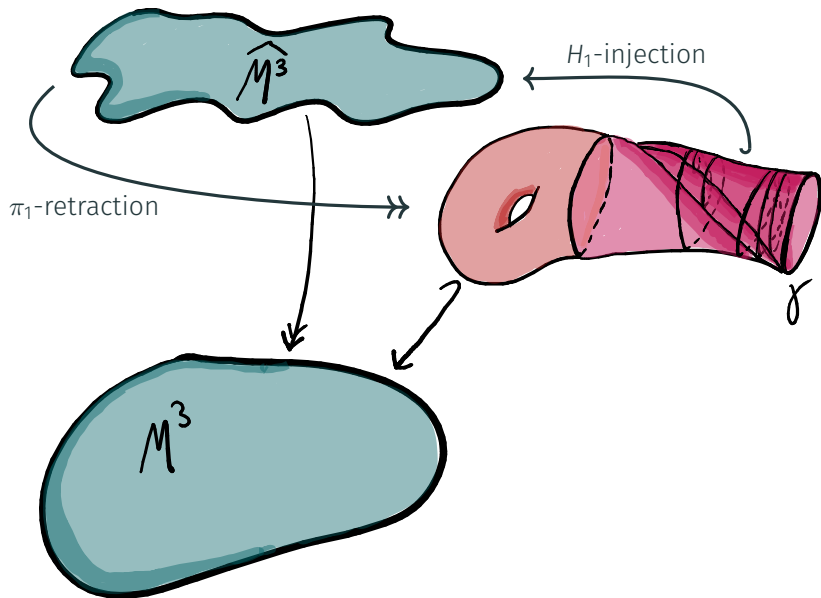
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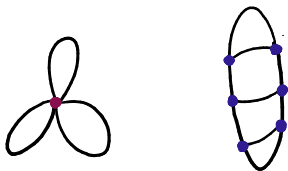


(Hyperbolic) Graphs of free groups with cyclic edges

(Fundamental groups of) graphs joined together by cylinders.

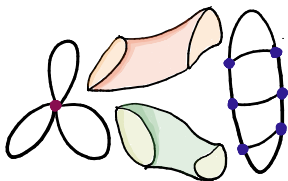
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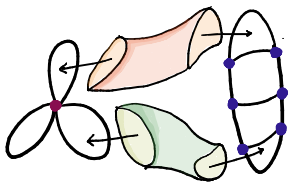
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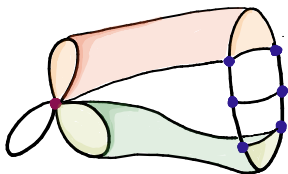
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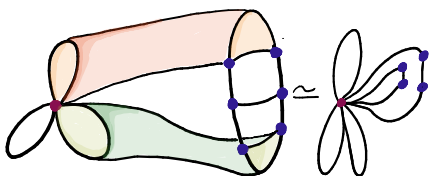
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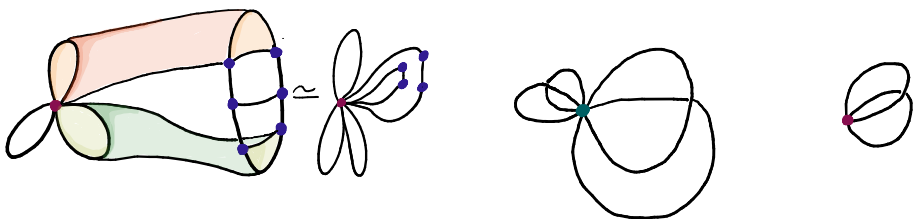
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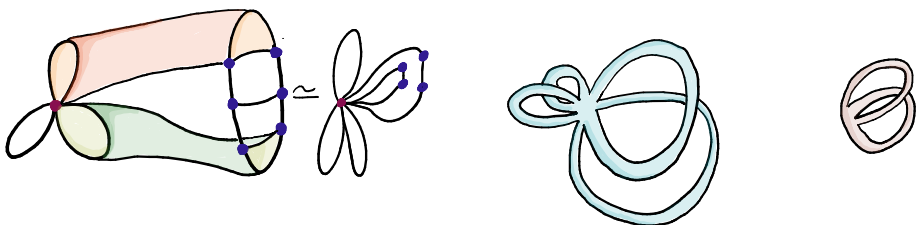
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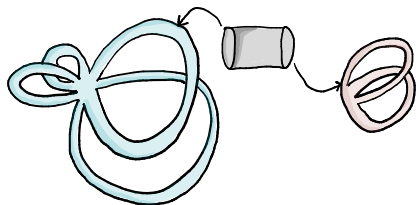
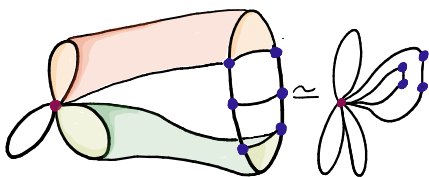
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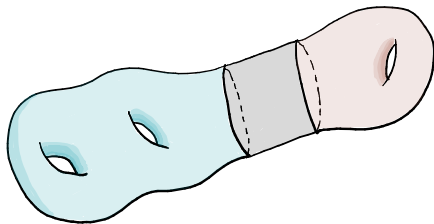
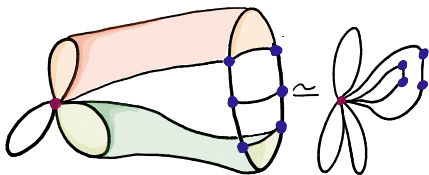
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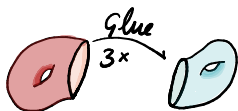
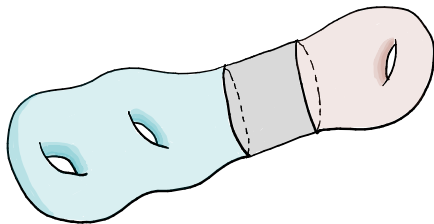
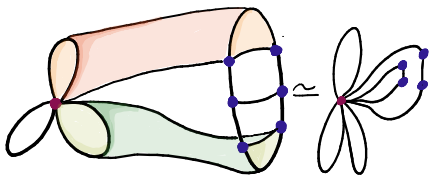
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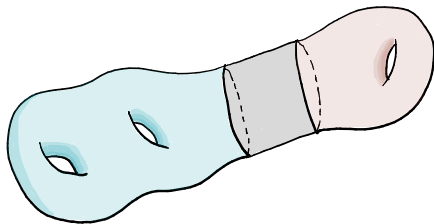
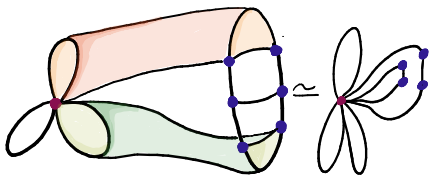
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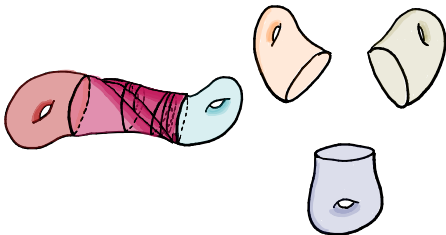
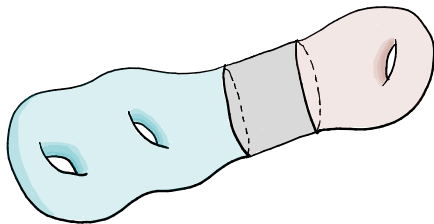
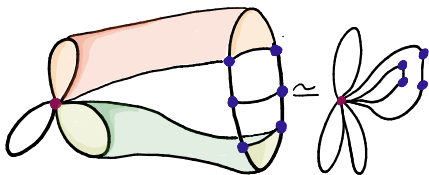
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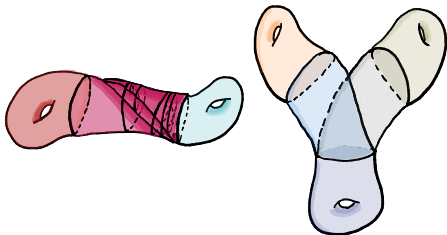
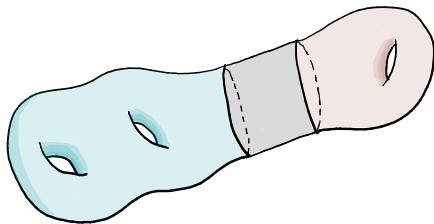
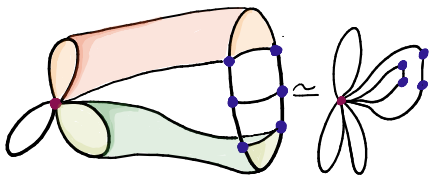
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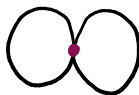
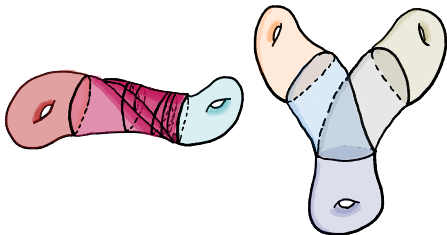
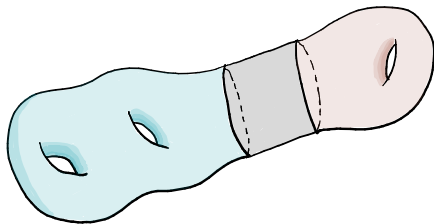
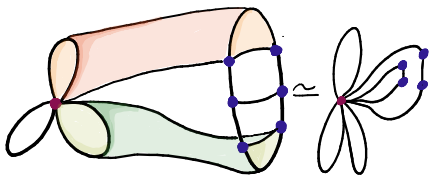
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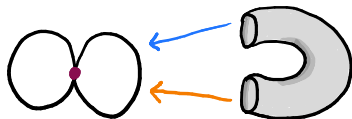
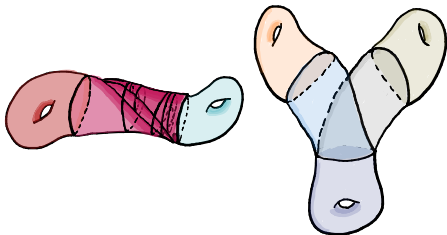
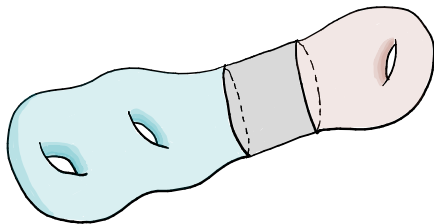
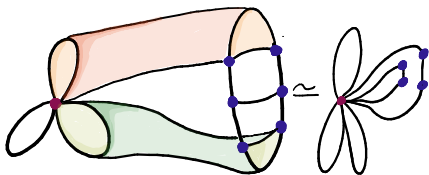
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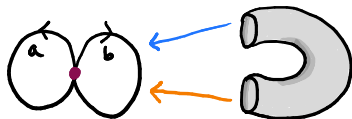
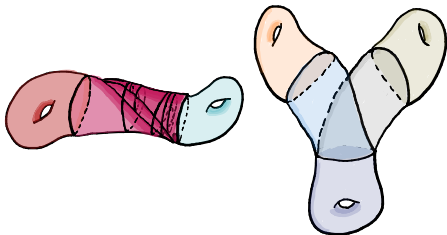
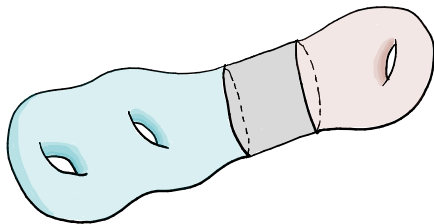
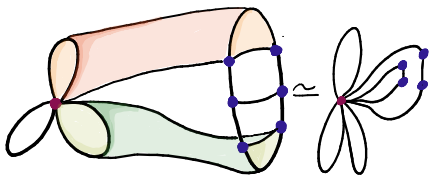
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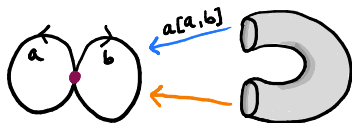
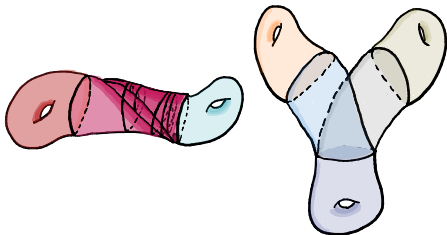
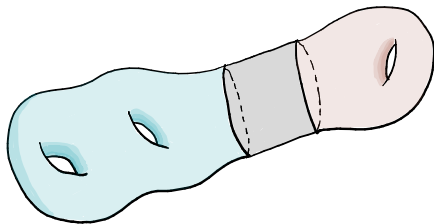
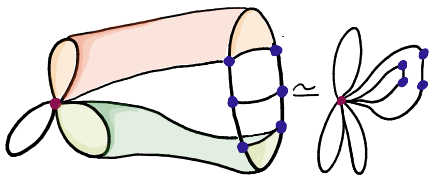
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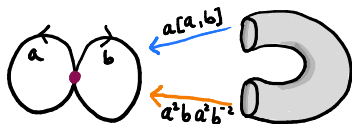
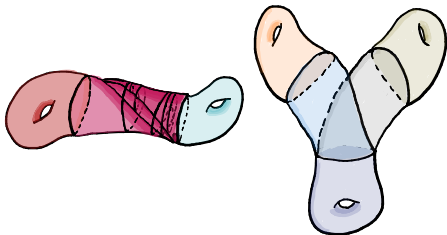
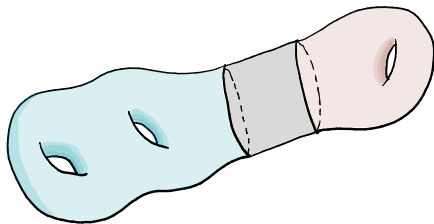
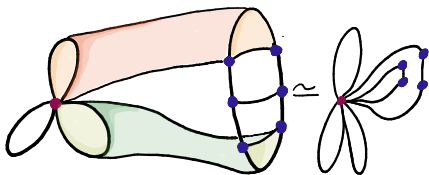
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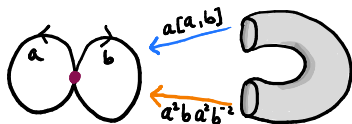
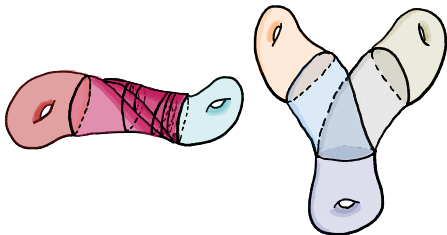
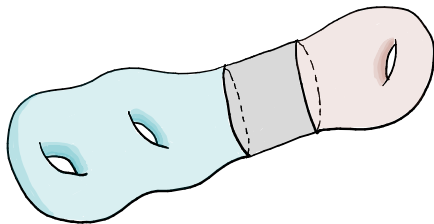
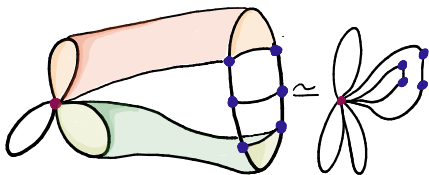
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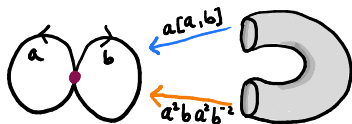
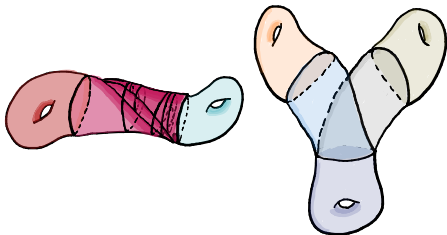
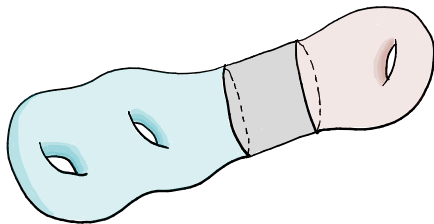
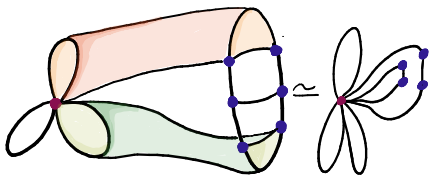
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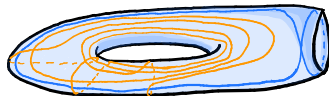
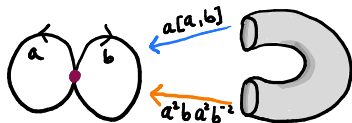
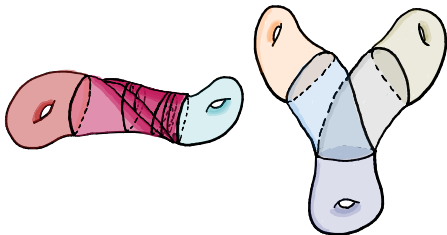
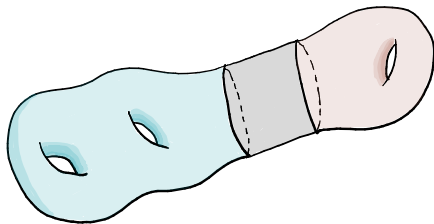
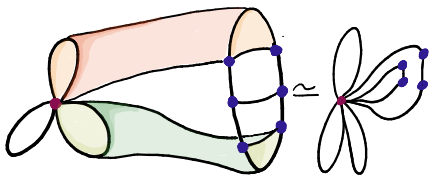
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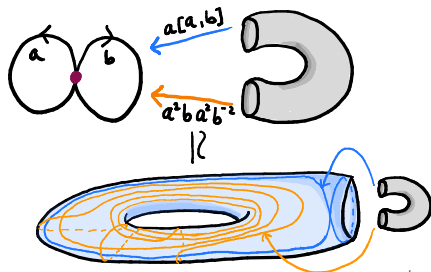
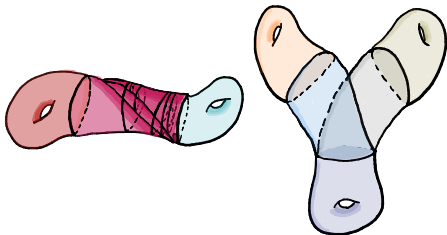
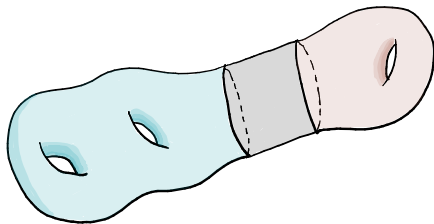
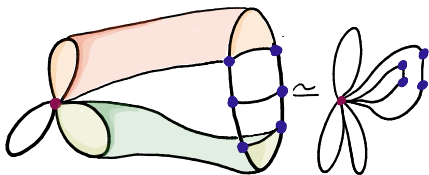
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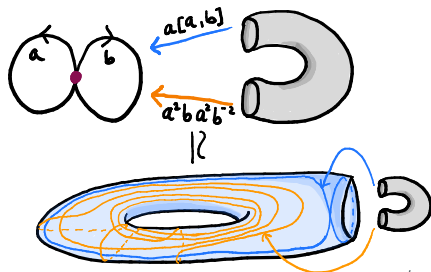
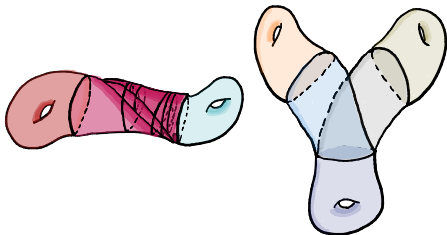
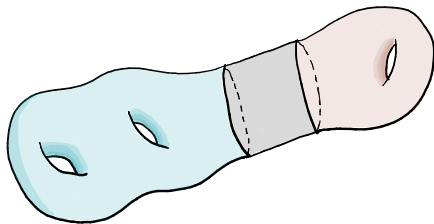
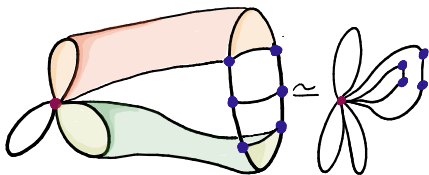
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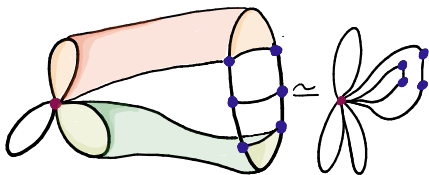
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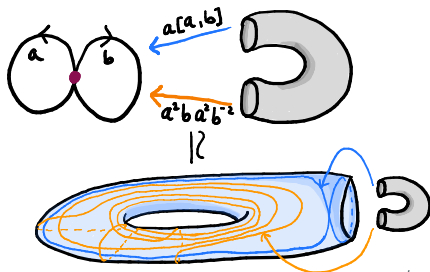
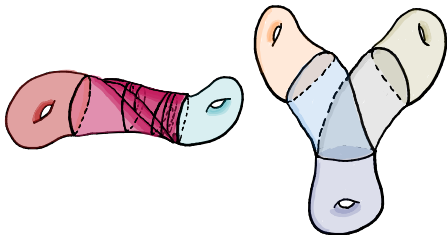
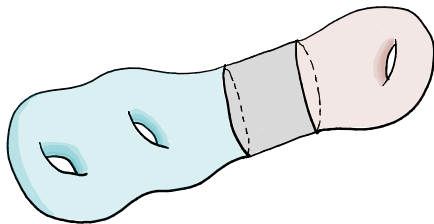


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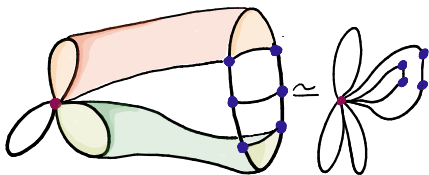


Free and surface groups
(and their free products)

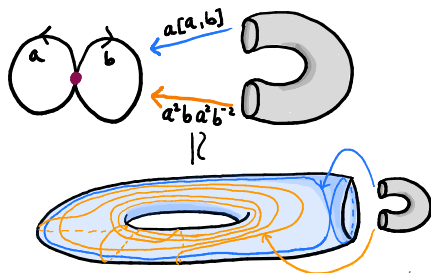
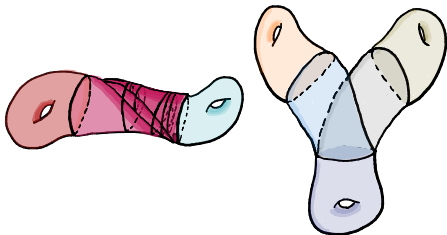
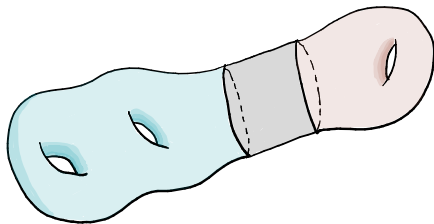


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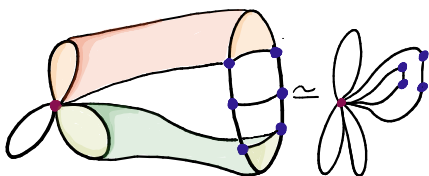


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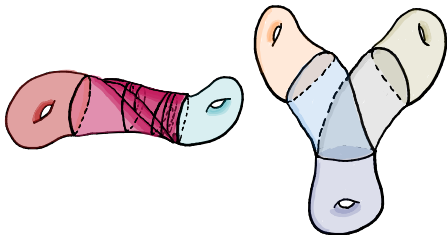
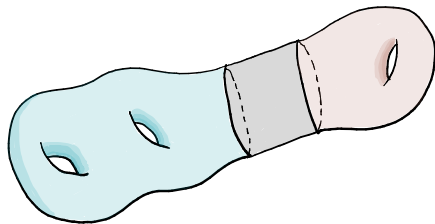


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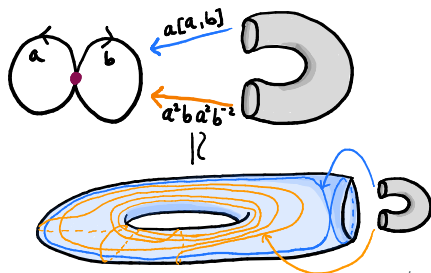
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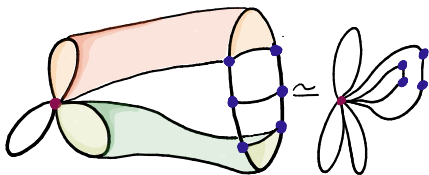


“More than a surface”

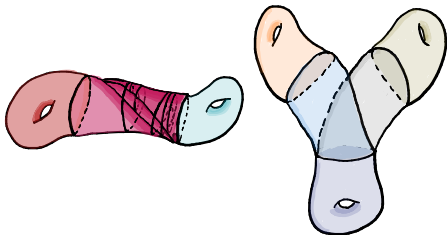
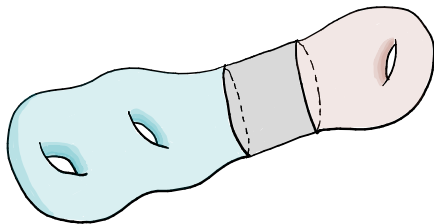


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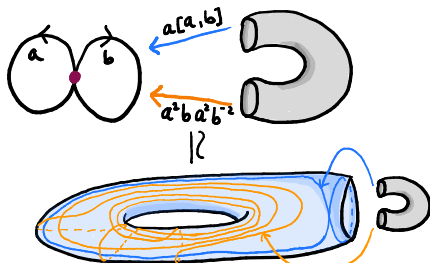
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Free and surface groups
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“More than a surface”



“Has a *rigid* vertex”

Homological torsion: Existence

Homological torsion: Existence

Theorem (Ascari-F '25)

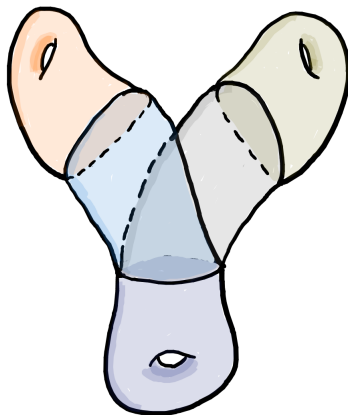
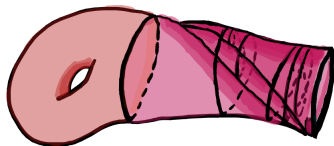
Let G be a hyperbolic graph of free groups with $\mathbb{Z}$ edges, which is not a free product of free and surface groups. Then G has abundant virtual homological torsion, that is,

for every finite abelian group A , there is a finite-index subgroup $G_0 \leq G$ such that

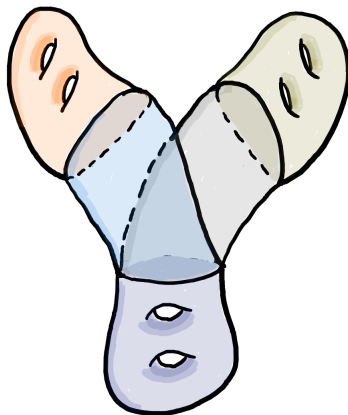
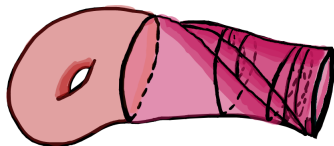
$$G_0^{\text{ab}} = A \oplus \cdots .$$

Homological torsion: By the book

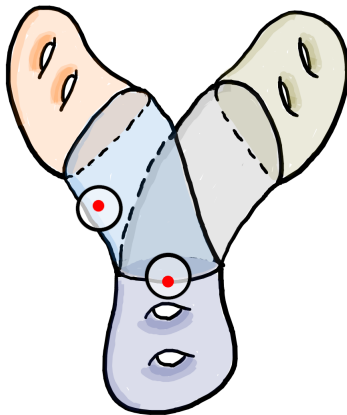
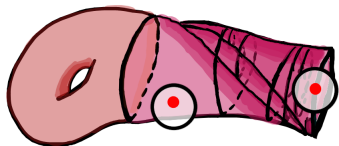
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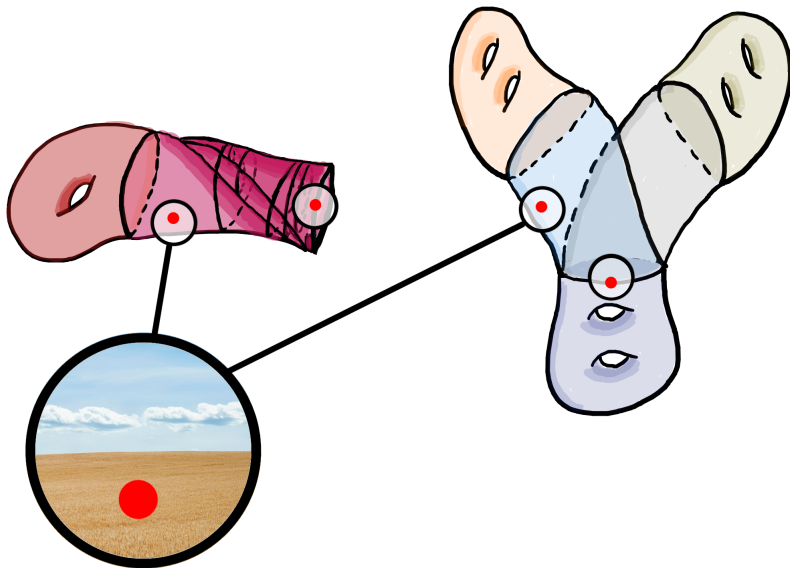
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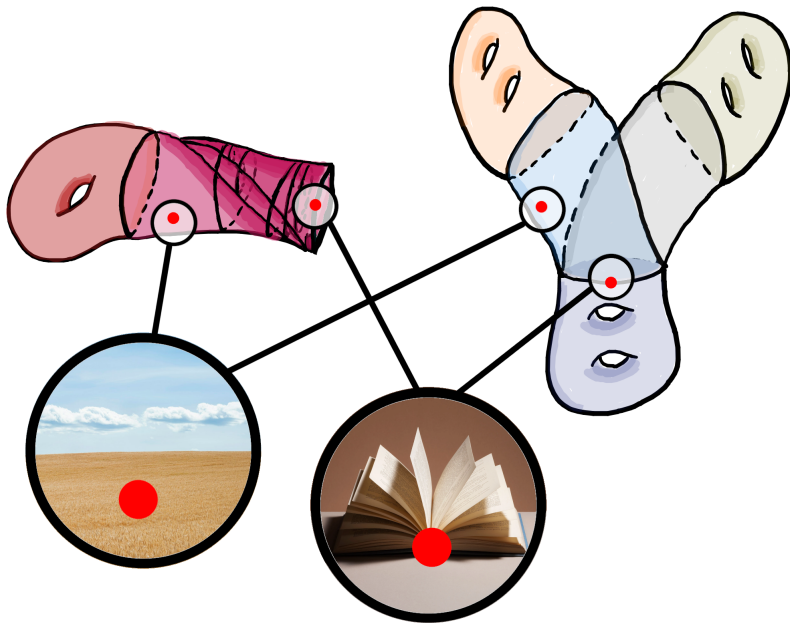
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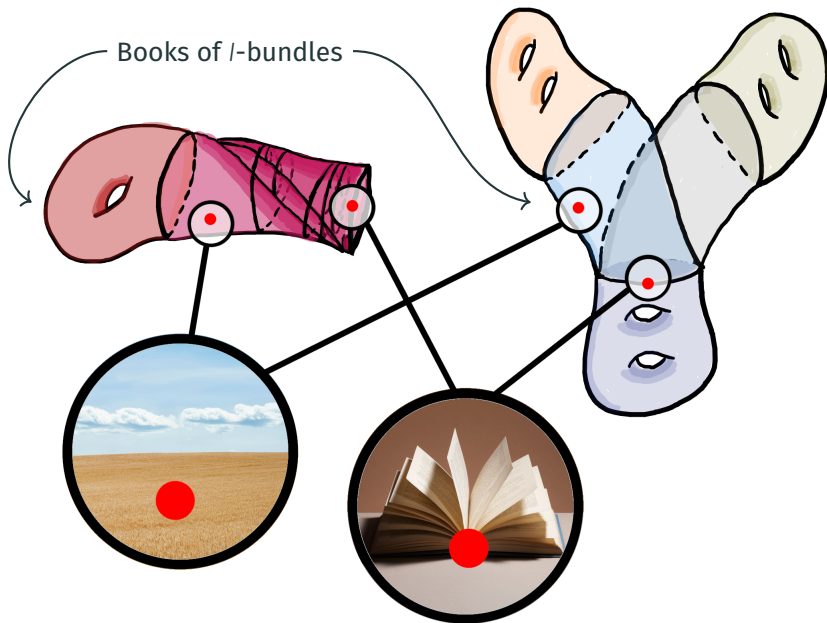
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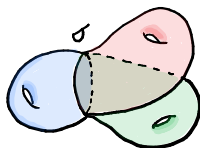


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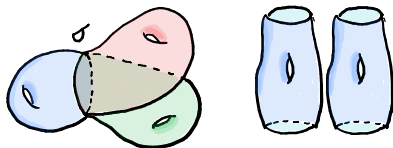


Homological torsion: Systems of surfaces

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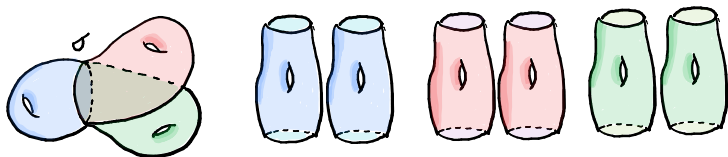
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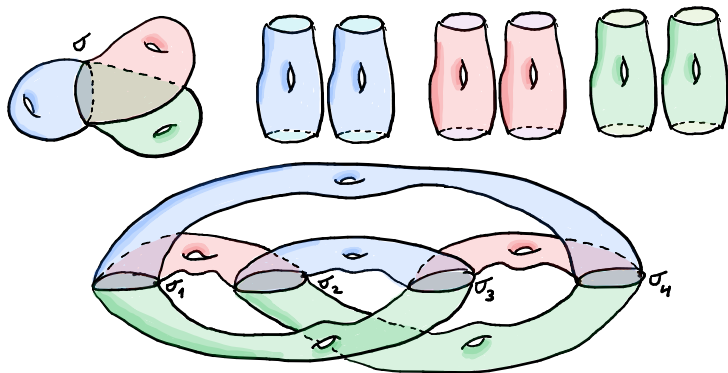
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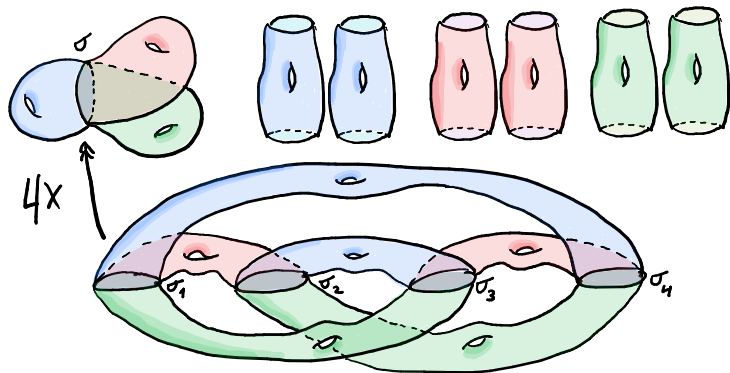
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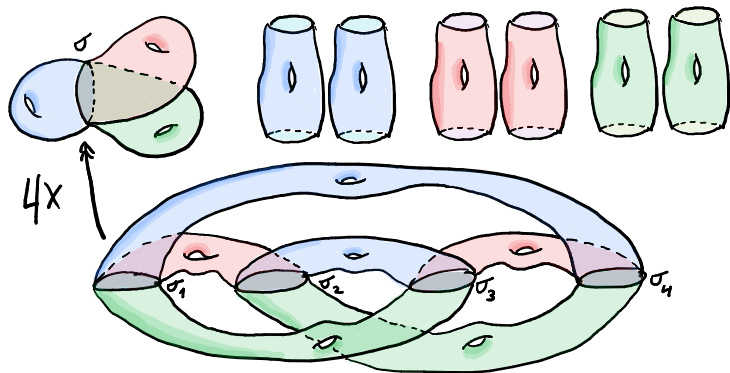
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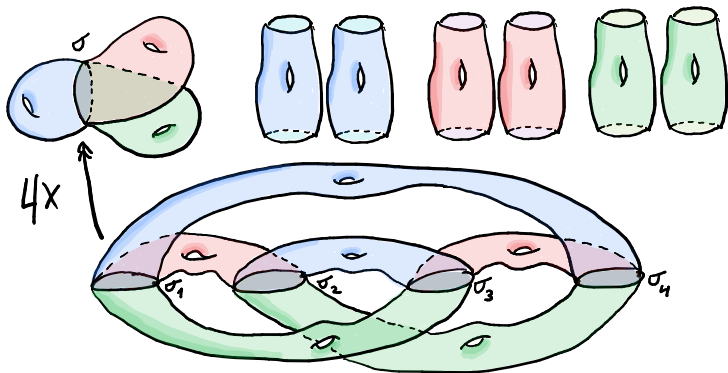


Homological torsion: Systems of surfaces



In first homology:

Homological torsion: Systems of surfaces



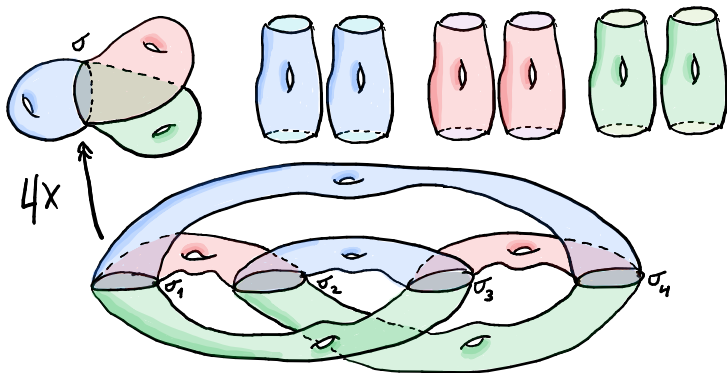
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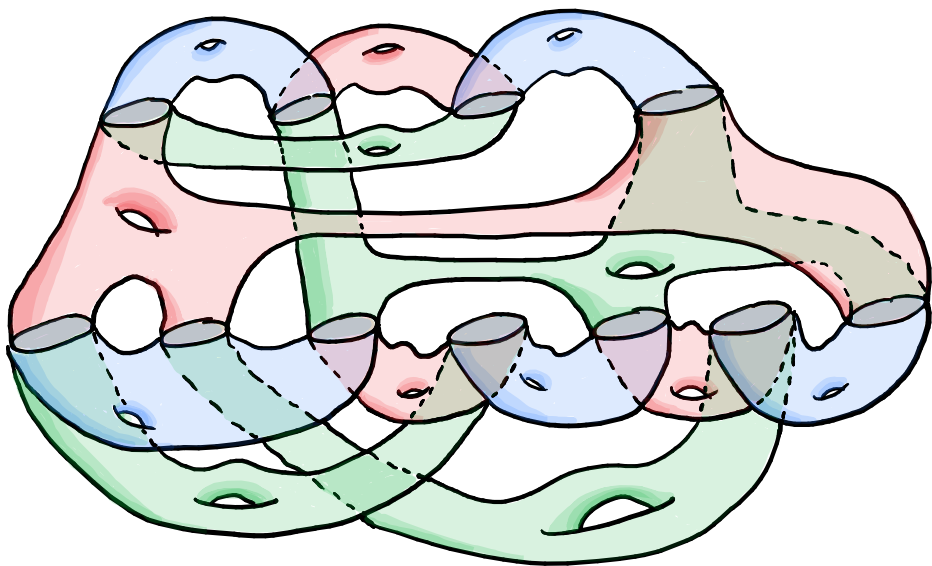
$\Rightarrow$

$$\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$$

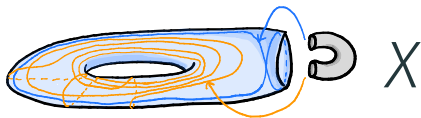
$$2 \cdot \sigma_1 = 0$$

Homological torsion: Two and three

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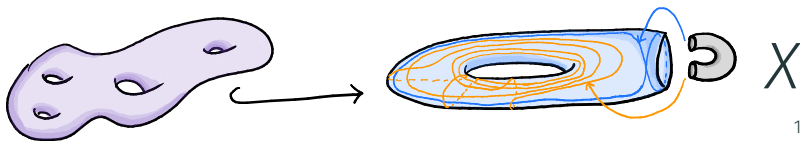
Enter a rigid vertex



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Theorem (Wilton '18)

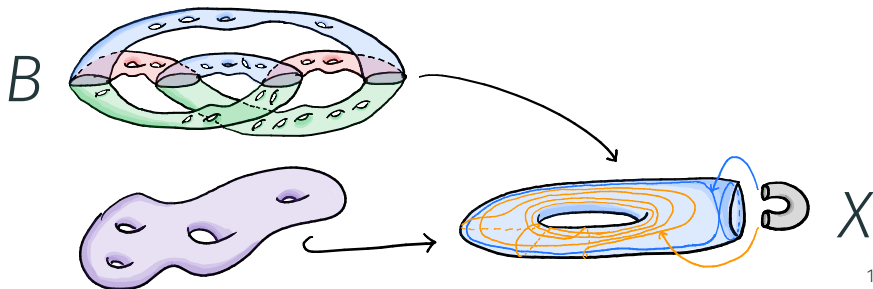
If G is not free then G has a surface subgroup.



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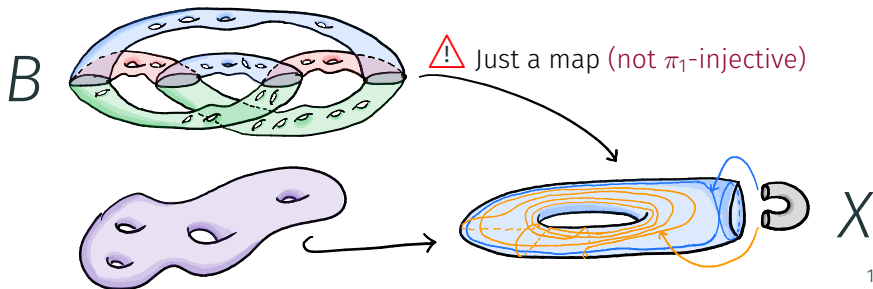
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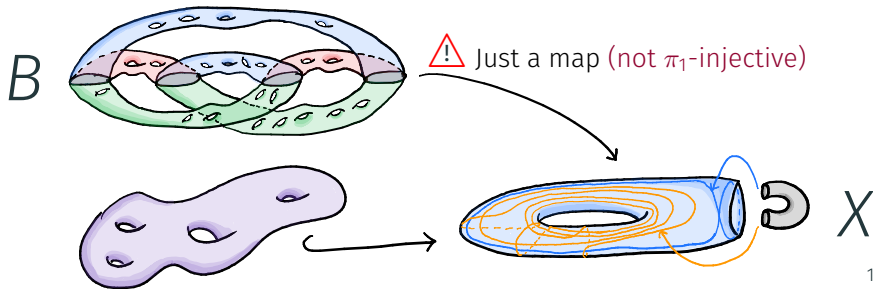
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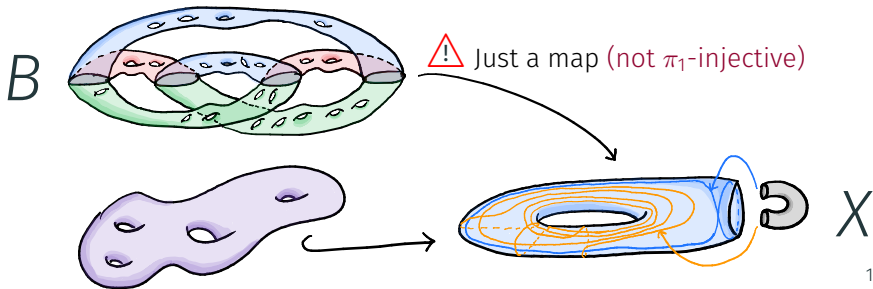
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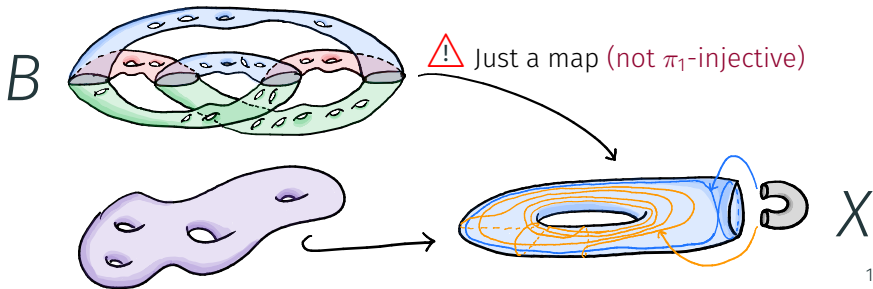
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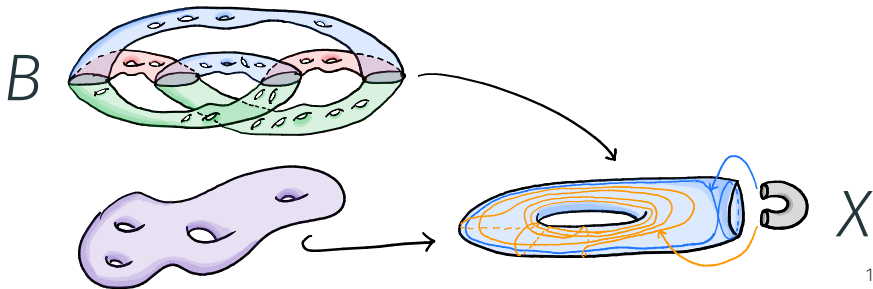
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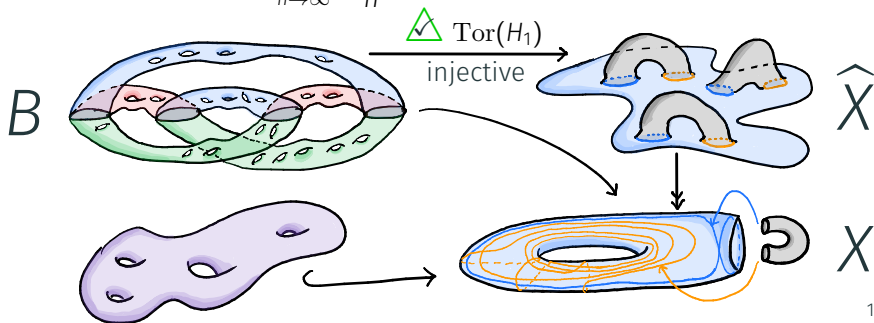
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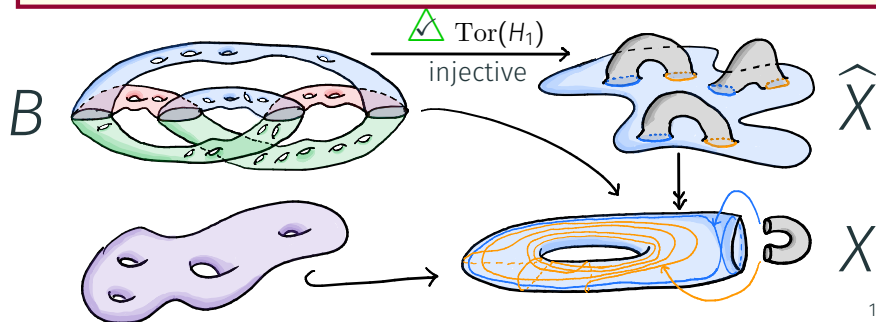
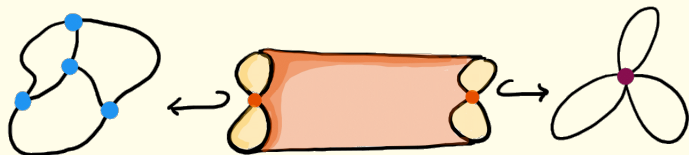
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Applicable to other settings?

e.g. graphs of graphs.



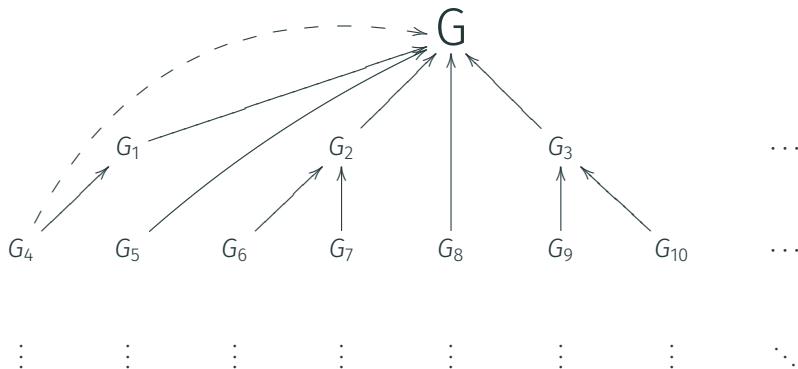
Profinite rigidity 101

Let G be a finitely generated *residually finite* group (every $g \in G$ survives in some finite quotient $q : G \twoheadrightarrow Q$).

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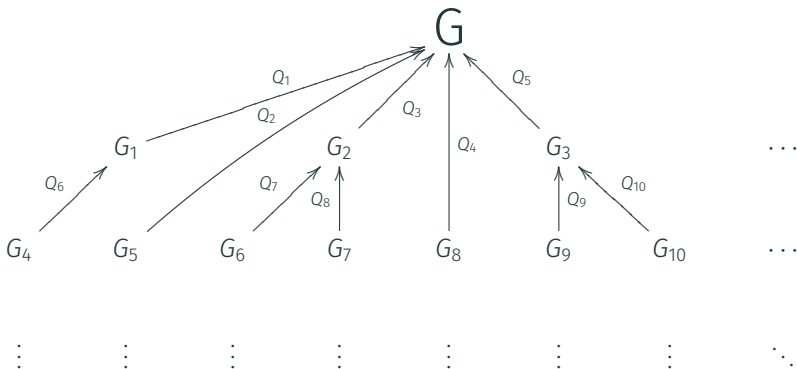
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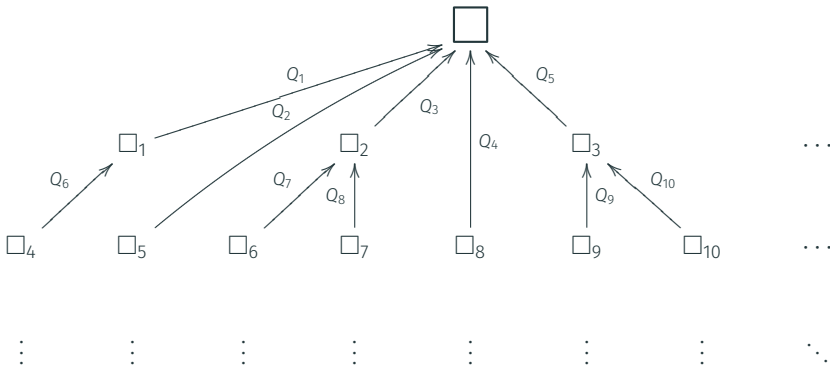
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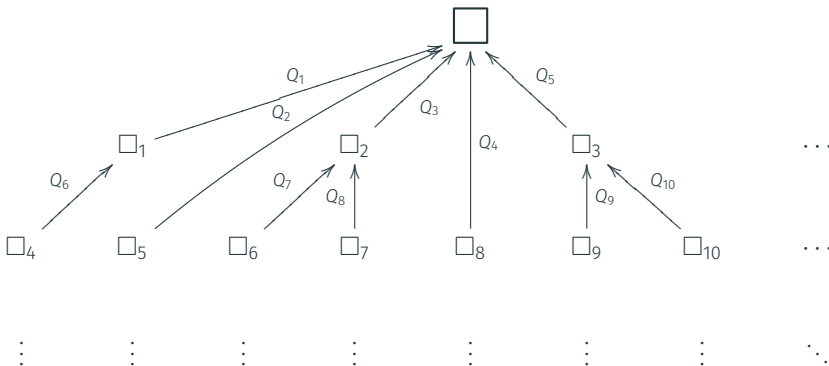
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Lattice/poset of finite-index normal subgroups of G



Question

Which fg residually finite groups H can fill the empty $\square$'s?

The basics

- Full lattice structure encoded in the set of finite quotients of G (“Dixon’s theorem”, DFPR ’82).

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Question (Remeslennikov)

Are finitely generated (non-abelian) free groups profinitely rigid?

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Theorem (Ascari-F '25)

Free products of free and surface groups are profinitely rigid amongst hyperbolic graphs of free groups with $\mathbb{Z}$ edges.

Growth: non-normal towers

Theorem (Ascari-F '25)

There exists a tower of finite-index non-normal subgroups

$$G \geq G_1 \geq G_2 \geq \cdots, \quad \text{with } \bigcap_n G_n = \{1\},$$

such that simultaneously for every prime p ,

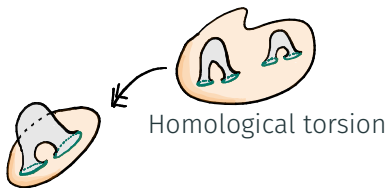
$$\lim_{n \rightarrow \infty} \frac{\log(|\mathrm{Tor}_p(G_n^{\mathrm{ab}})|)}{[G : G_n]} > 0.$$

Fractal-like towers

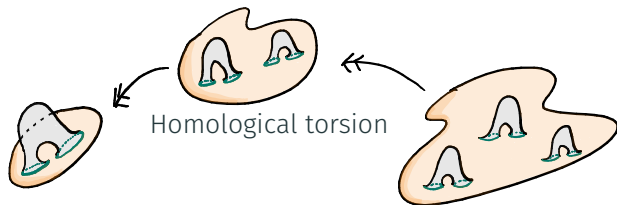
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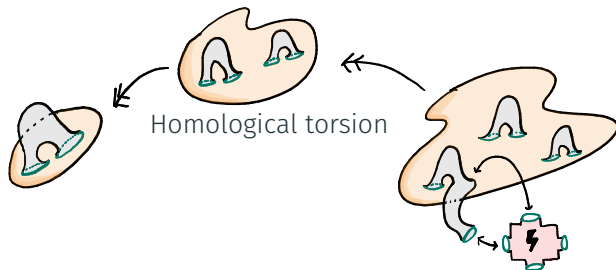
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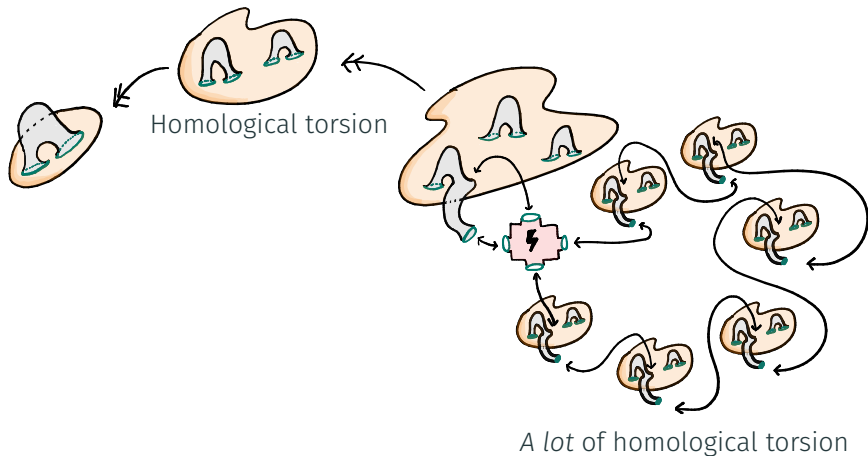
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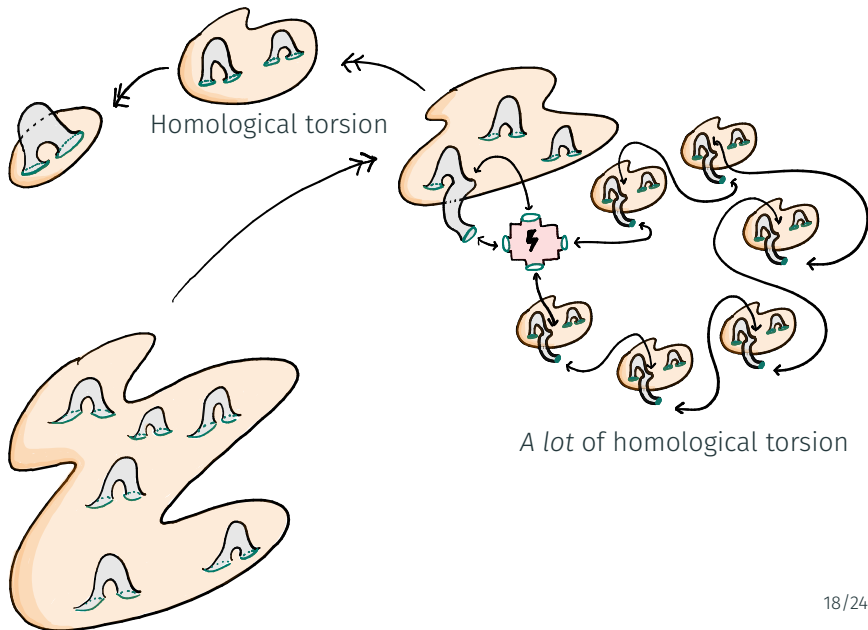
Fractal-like towers



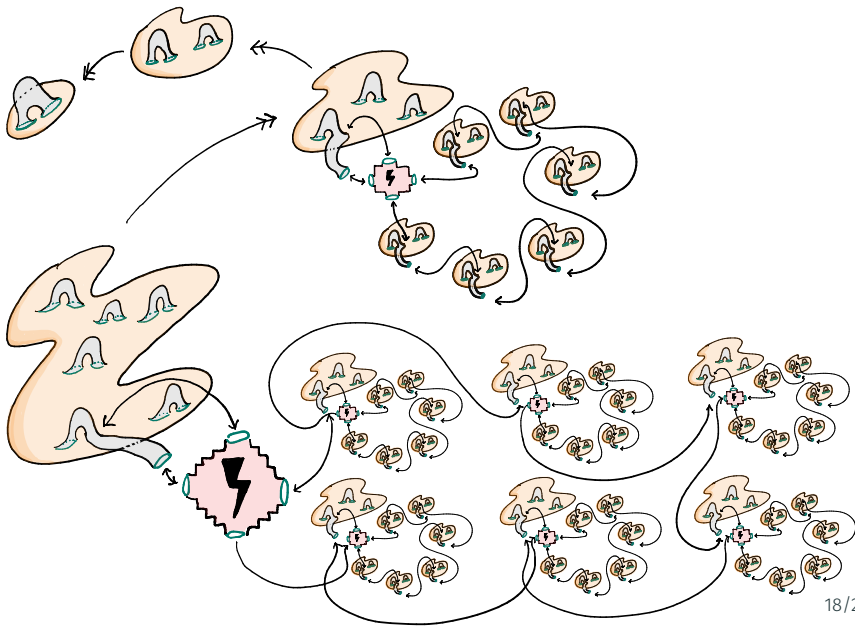
Fractal-like towers



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Fractal-like towers



Growth: normal towers

Theorem (F '25)

In books of I -bundles, homological torsion grows subexponentially along any exhausting tower of finite-index normal subgroups.

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No infinite amenable normal subgroups

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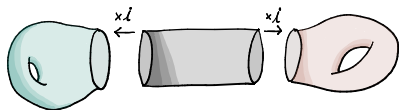
In books of I -bundles, homological torsion grows subexponentially along any exhausting tower of finite-index normal subgroups.

- **Tension:** Books of I -bundles can have abundant virtual homological torsion.
- Bergeron-Venkatesh: such behaviour expected in dimension ≥ 4 .

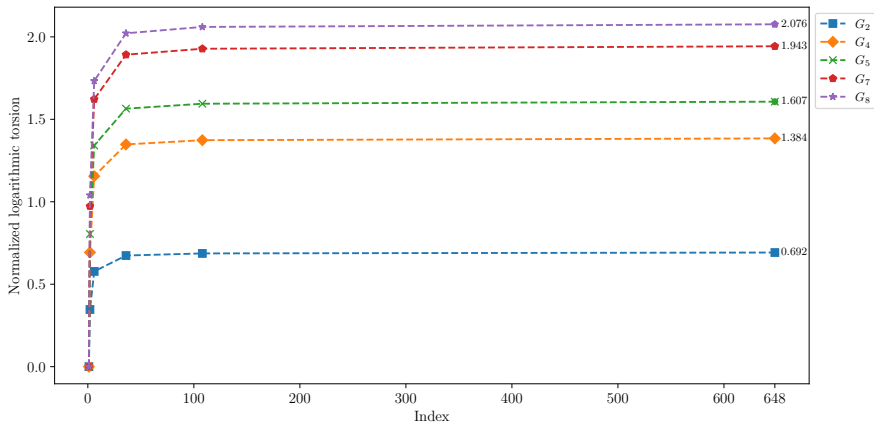
A computation: unwrapping cylinders

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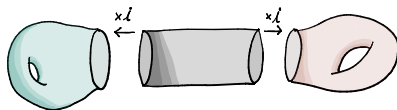
$$G_i = \langle a, b \rangle_{[a,b]^i=[c,d]^i} \langle c, d \rangle$$



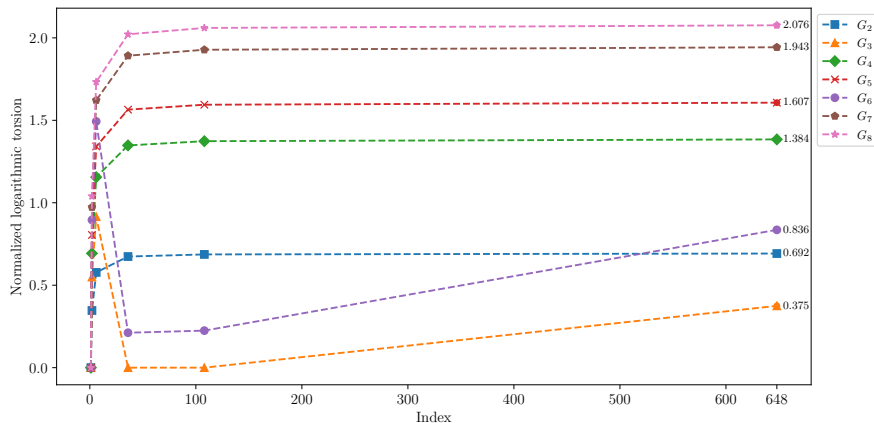
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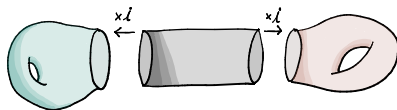
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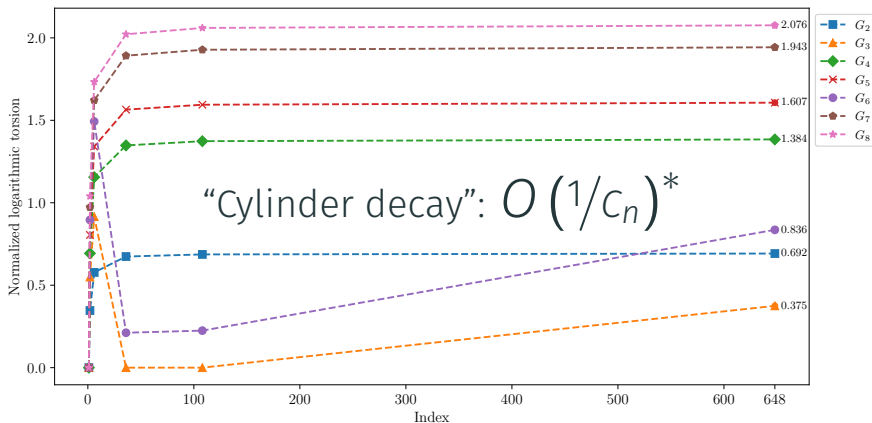
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A computation: unwrapping cylinders



* C_n measures the wrapping degree of the cylinder in the n^{th} cover.

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Enter a rigid vertex: revisited

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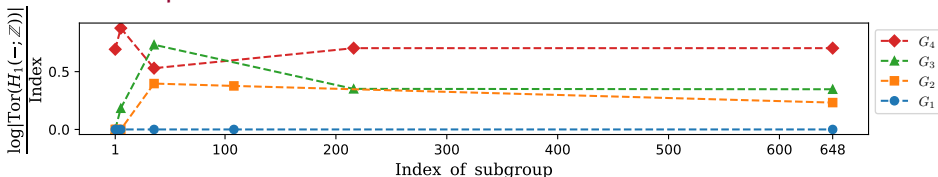
-   **Restrictions lifted:** Torsion no longer confined to cylinders.

Enter a rigid vertex: revisited

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- Refining the construction  torsion elements **inside** a rigid vertex. **Multiply fast, difficult to control.**

Enter a rigid vertex: revisited

- 🔓👮 **Restrictions lifted:** Torsion no longer confined to cylinders.
- Refining the construction $\rightsquigarrow$ torsion elements **inside** a rigid vertex. **Multiply fast, difficult to control.**
- Computations:**



$$G_1 = \pi_1 \left(\text{Diagram of a genus-1 surface} \right)$$

G_{i+1} obtained from G_i by adding a single letter to its presentation

Enter a rigid vertex: revisited II

Theorem (F '26)

If G has a rigid vertex then

$$\frac{\log(|\mathrm{Tor}(G_n^{\mathrm{ab}})|)}{[G : G_n]} = O(\log(C_n)/C_n) \xrightarrow{n \rightarrow \infty} 0$$

where C_n measures the wrapping degree of cylinders in the n -th cover.

The landscape

The landscape

$\text{Tor}(H_1)$	(Hyp) 3-manifolds	Surface groups	Books of I -bundles	Rigid vertex
Abundance	 (Sun '15)		 (Ascari-F '25)	 (Ascari-F '25)
Non-normal exponential growth	 (Liu '19)		 (Ascari-F '25)	 (Ascari-F '25)
Normal growth rate	 (Exp.)		 Subexp. (F '25)	 Subexp. (F, '26)

The landscape

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Normal growth rate	 (Exp.)		 Subexp. (F '25)	 Subexp. (F, '26)
"Cylinder decay" rate			$O(1/C_n)$	$O(\log(C_n)/C_n)$

What's next?

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$\rightsquigarrow$ cross the threshold?



Thank you for listening!

